Astronaut Training

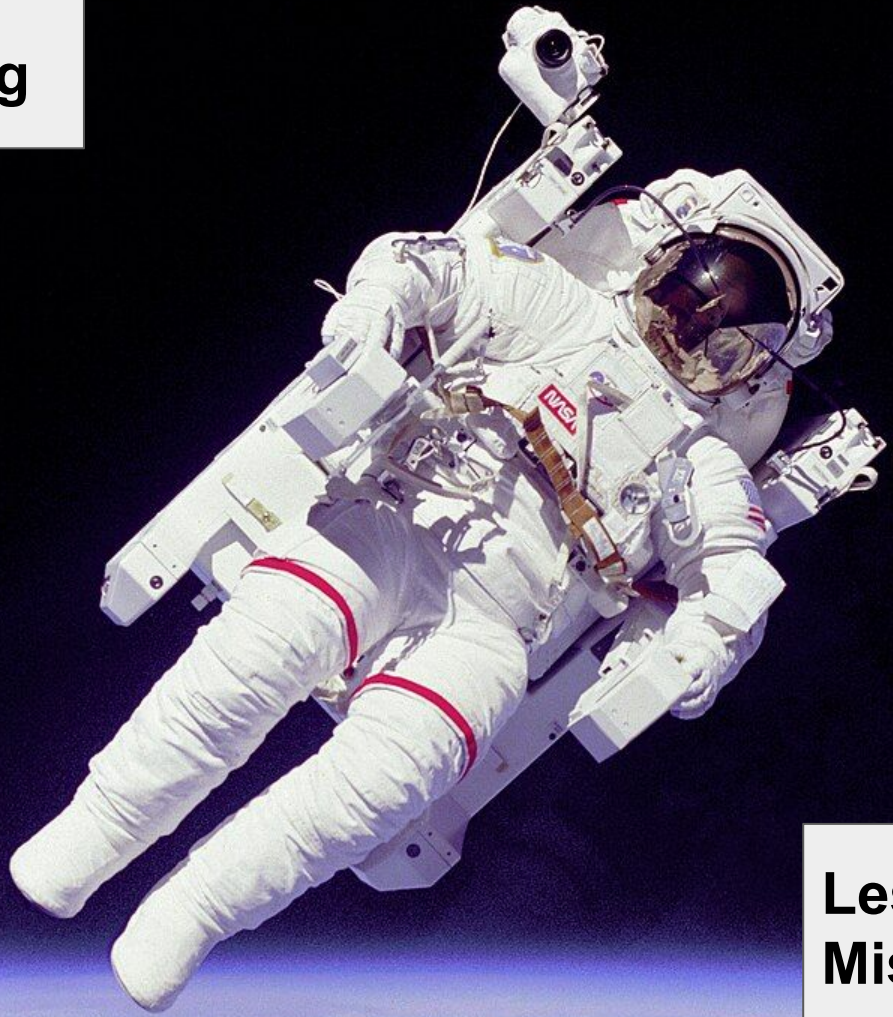


Image credit: NASA

Lesson 1: Mission Briefing

Welcome to your Astronaut Training!

- Lesson 1: Mission Briefing
- Lesson 2: Astronaut Selection & Training Facilities
- Lesson 3: Living in Space
- Lesson 4: Spacewalks and Spacesuits
- Lesson 5: The Work of Astronauts
- Lesson 6: Mars and the Future of Space Exploration

My name is Sven
I am your teacher for this class!

- I grew up in Switzerland.
- I love cheese and chocolate.
- I am dyslexic.
- My favourite hobby is traveling.
- My son Finn was born last Christmas!
- I already live for nearly 2 years here in Austin.







How the classes work

Red Boxes Show Activities

We will do a hands on activity!

Green Boxes are questions

Raise your hand if you want to answer!

Blue Boxes are fun facts or answers

Here you learn something new!

- Raise your hand if you have a question.
- You are always free to ask your neighbour.
- Be respectful of others. Give them time and space to participate as well.
- Pay attention to the lecture. You will learn some really cool stuff!

What is an Astronaut?

- A person whose ***profession*** is to travel ***beyond the Earth's atmosphere***
  - A pilots are professionals, but they stay within Earth's atmosphere
  - A space tourist travels to space but does not work there
- The first human in space was the Soviet pilot Yuri Gagarin in 1961







Let’s introduce ourselves!

Group Activity: Introductions

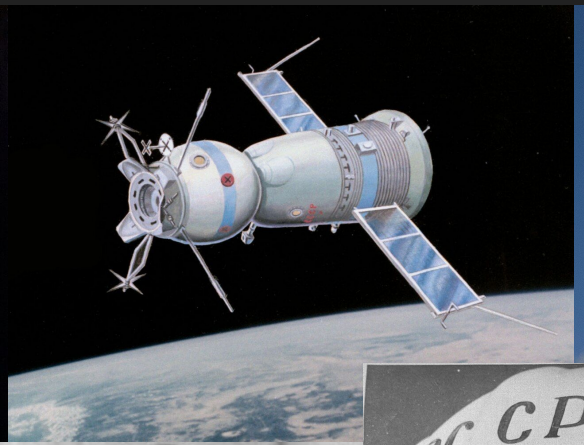
We go through the room:

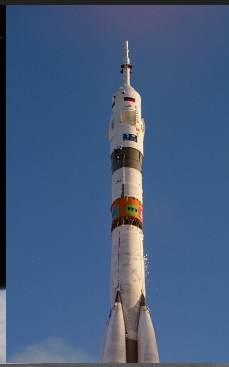
- Say your name
- Say something you like about astronauts or space

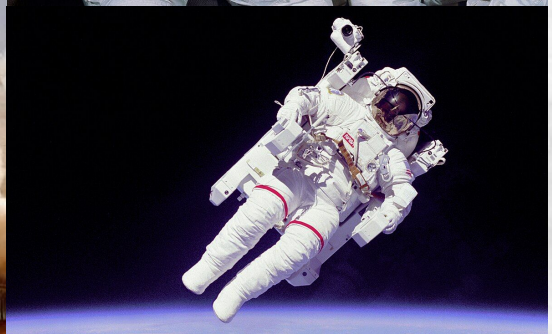
Today's Topic: Mission Briefing

















Different roles of astronauts

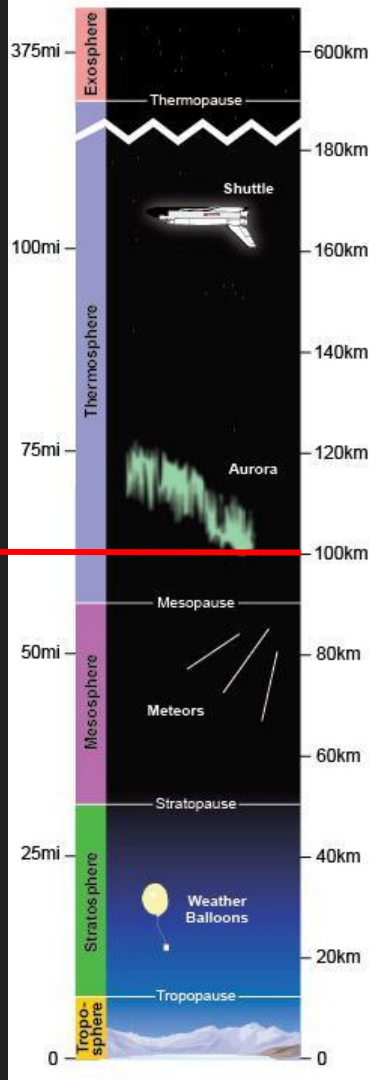
What do you think are different roles of astronauts?

- **Pilot:** the astronaut who flies the spacecraft
  - Typically jet pilots by education
  - Also: command module, lunar module, and docking module pilot
- **Commander:** responsible for mission success and safety
  - Also: payload, and joint operations commander
- **System operator:** Manages the spacecraft during flight
- **Mission Specialist:** An astronaut with a mission-specific duty
  - They do not need to be NASA trained astronauts
  - Can be: scientists, doctors, educators, engineers

Where Does Space Start?

Activity: Where does space start?
I go through the atmosphere and you raise your hand if you think space starts here

- The official definition is that space starts at the Kármán line at 62 miles (or 100km)
- This is an arbitrary definition and there is no clear boundary between Earth and space
- Anyone working above that line is considered an Astronaut



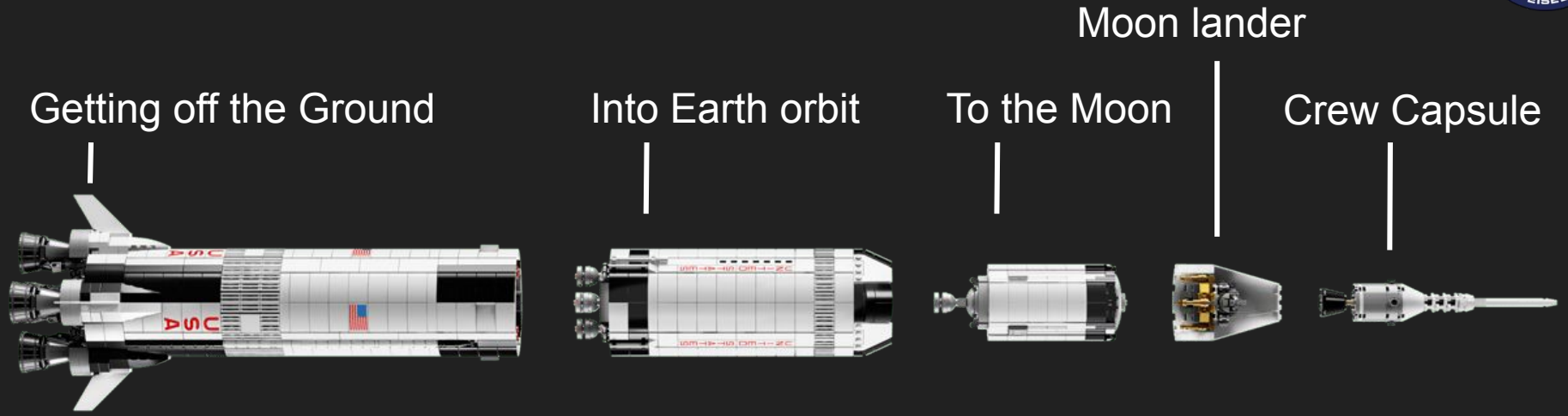
How to reach space

- Rockets are objects that turn fuel into gas and release that gas in order to move
- Rockets move because of Newton’s third law of motion:
  - For every action there is an equal and opposite reaction
- The gas made from the fuel is pushed out the back of the rocket which makes the rocket move up!




Action
Reaction

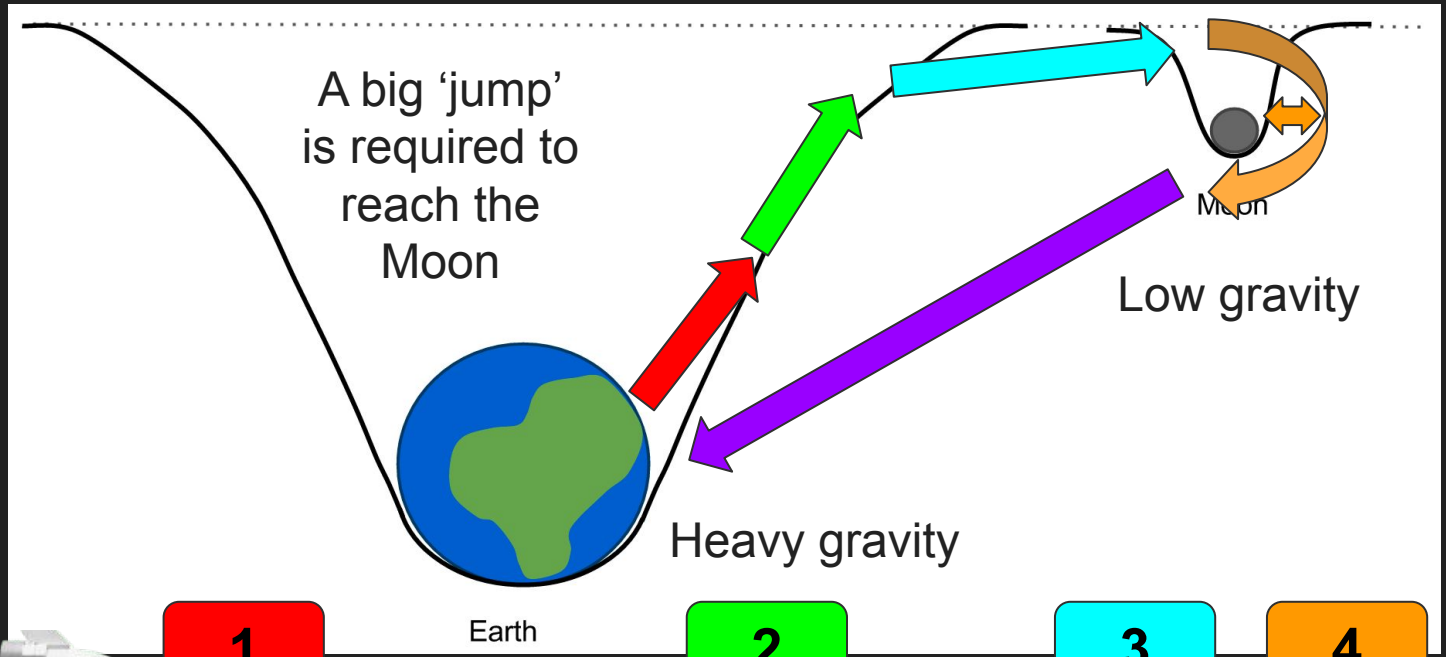
Most famous rocket: The Saturn V



- The first Saturn V was used to bring the first humans to the moon.
- 24 astronauts flew with the Saturn V
- The rightmost stage is the first, the leftmost is the last.
- The image shown here is from a LEGO set!



Saturn V took Astronauts to the Moon



Coming back from the moon only requires a ‘small’ jump

The Space Shuttle

- The space shuttles are reusable rockets.
- There were 5 space shuttles:

  Columbia, Challenger, Discovery, Atlantis, Endeavour
- In total they flew **135 Missions**:
  - Deployed new satellites into space
  - Repaired the Hubble Space Telescope
  - Flew to the ISS
  - 8 classified missions for the Military
- **355 people from 16 countries** flew on the space shuttles





The Soyuz Program

- The first Soyuz rocket flew in 1967
- The Soyuz program is still flying today
- They completed **157 crewd rocket launches** to date
- Only 4 of those crewed missions failed

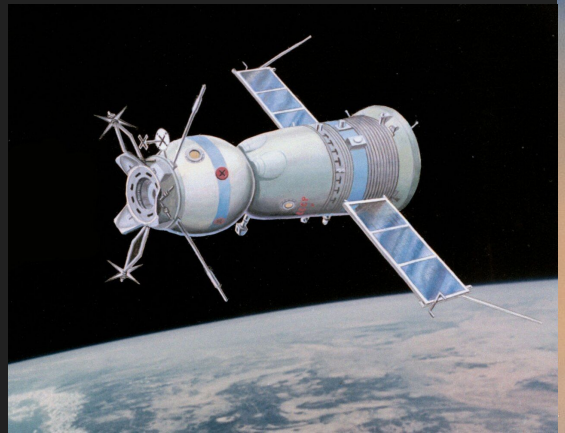




Image credit: NASA

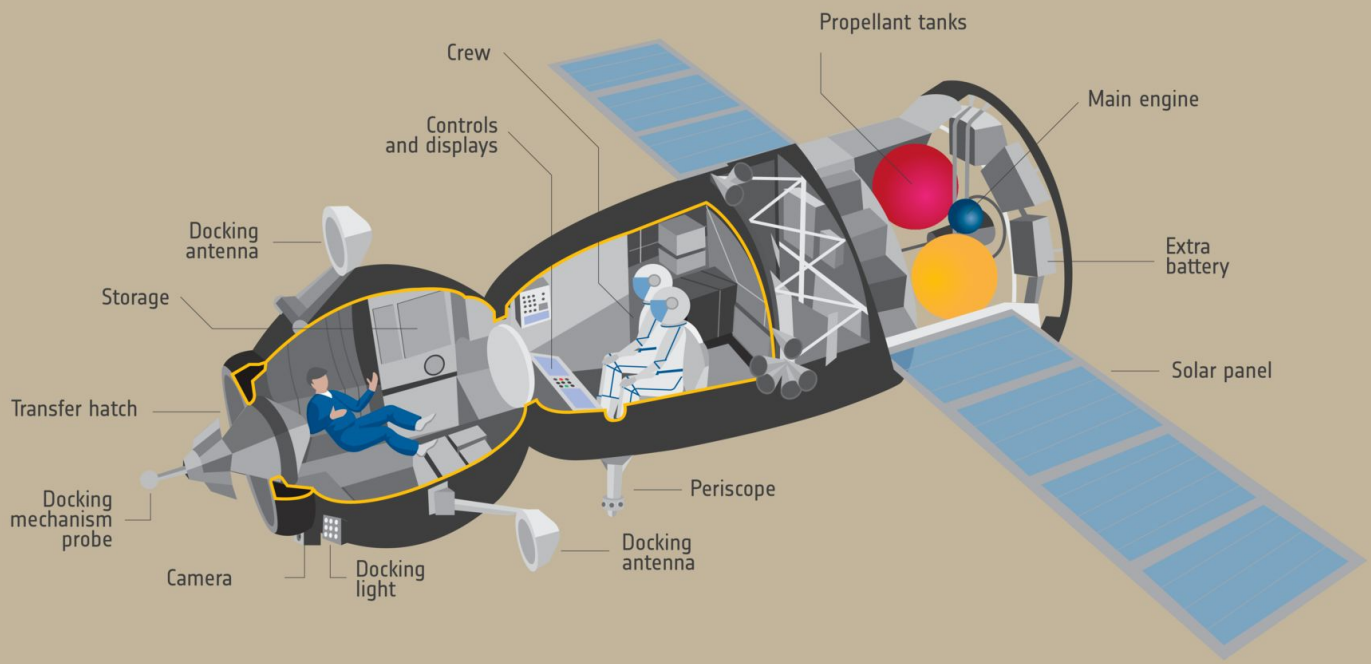


Image credit: ESA

SPECIFICATIONS

Habitable volume: 10 m^3

Maximum diameter: 2.72 m

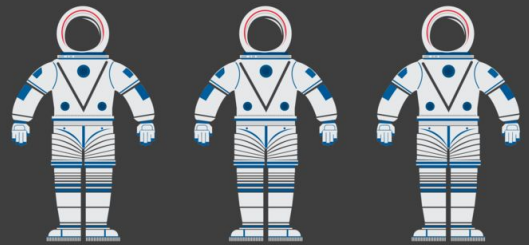
Main engine thrust: 2942 N

Engine propellant: N2O4 / UDMH

Electrical system: Solar panels and batteries

Solar panel power: 0.60 kW average

CREW: three astronauts

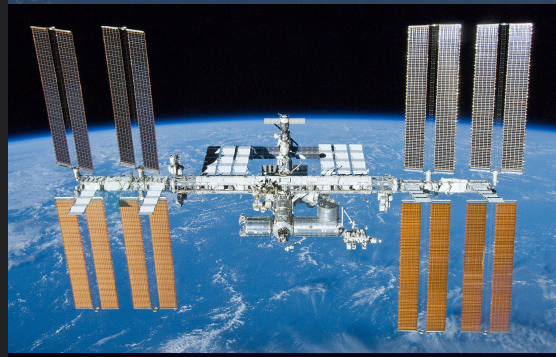




Notable Space Missions

- **The Apollo** program is the only one that flew to the Moon. They flew 21 Astronauts to the Moon.
- **The International Space Station** has up to 7 Astronauts. In total 290+ people from 29 Countries have lived there.
  - 355 people from 16 countries flew on space shuttles
- **The Soyuz** program has brought the most humans into space. A total rough 400 people (including 3 tourists)
- A total of 637 people from 47 Countries have flown to space







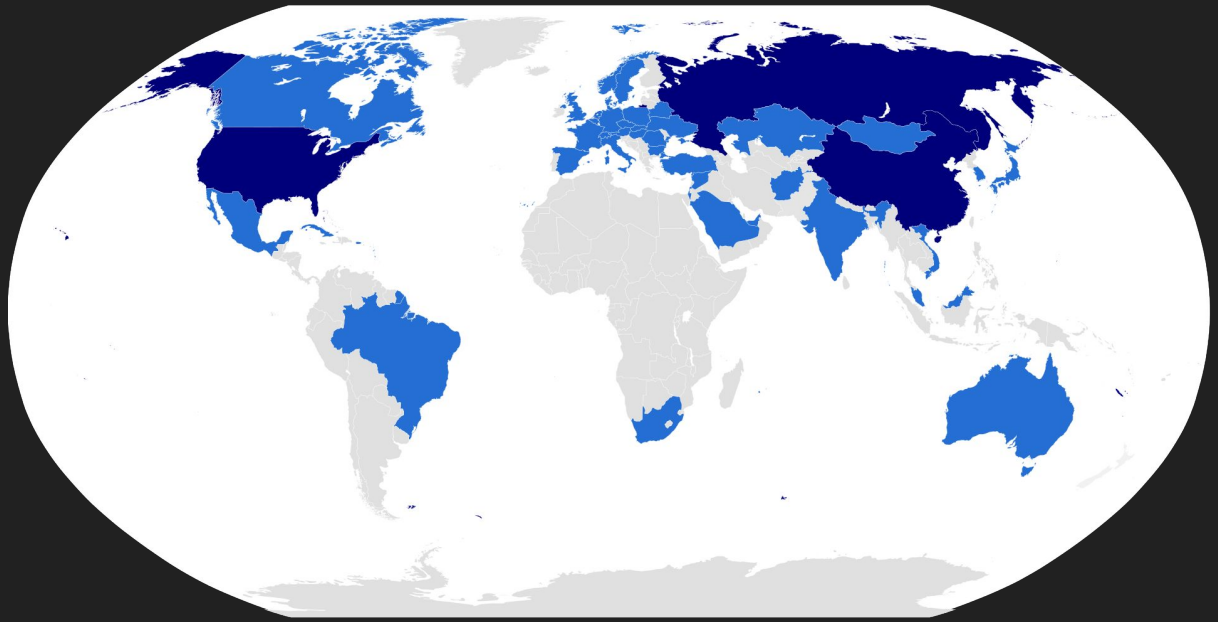
Who has been to space?

People from which countries have flown to space?

- Afghanistan
- Australia
- Belarus
- Brazil
- Bulgaria
- Canada
- China
- Cuba
- Czech.
- Austria
- Belgium
- Denmark
- France
- Germany
- Hungary
- Italy
- Netherlands
- Norway
- Malta
- Poland
- Romania
- Spain
- Sweden
- CH
- UK
- India
- Israel
- Japan
- Kazakhstan
- Malaysia
- Mexico
- Mongolia
- Russia
- Saudi Arabia
- Slovakia
- South Africa
- South Korea
- Soviet Union
- Syria
- Turkey
- Ukraine
- UAE
- USA
- Vietnam

Who has been to space?

- These Countries have all traveled to space



- Countries that have their own human spacecrafts:
  - USA
  - China
  - Russia

- 637 Humans have traveled to space:
  - 379 Americans
  - 72 Soviets
  - 56 Russians
  - 30 Chinese
  - 107 others

Famous Astronauts



Claudie Nicollier
First and only Swiss astronaut

Michael Collins
Module Pilot

Neil Armstrong
Commander

Buzz Aldrin
Lunar Pilot

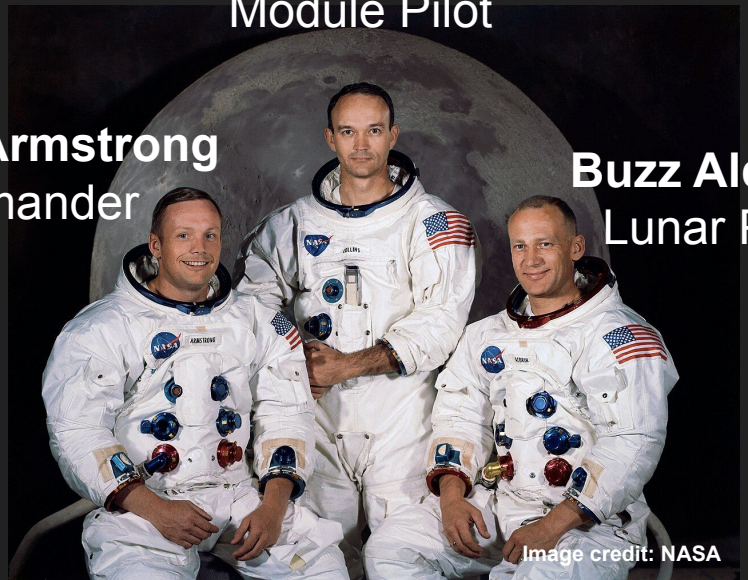


Image credit: NASA

Valentina Tereshkova
First Woman in space (1963)

Yuri Gagarin
Roscosmos (1961)



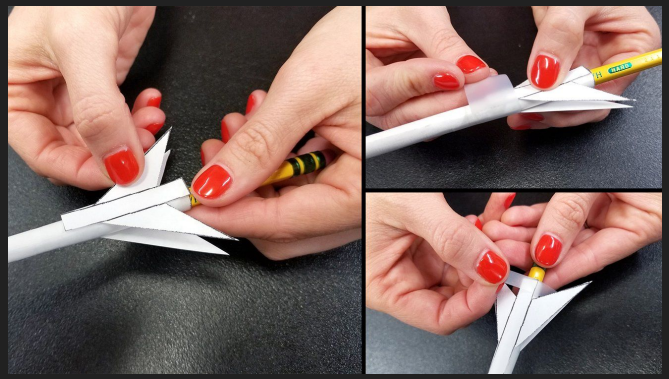


Image credit: Sovfot

Space Flight accidents

- A total of 19 people died during space flights
  - Starting and landing is most dangerous
  - Only 3 people died in space
- A total of 11 people died during preparations for space flights
- There are 40+ major non-lethal accidents related to space flight
- Space agencies take all of these accidents very seriously
- **No astronaut has died during space flight (launching, flying or landing) since 2003!**

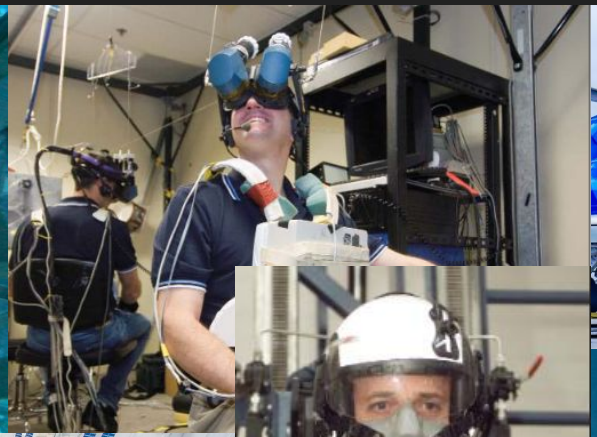
Today's Activity: Launch your rocket!



- Split into groups
- Take some paper, fins, tap and straws.
- Build a paper rocket for the your straw
- Make sure its tight and aerodynamic!
- Fly your rocket!

Next Week: Astronaut selection














XXII
BECOMING
ASTRONAUTS
ARE YOU NEXT?